

Kan Zhu

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About Me

I am Kan Zhu, a fourth year PhD student at University of Washington's Paul G. Allen School of Computer Science and Engineering, co-advised by [Baris Kasikci](#) and [Arvind Krishnamurthy](#).

Research Interests

I build systems that make large language model (LLM) inference faster and cheaper. My work spans the inference stack: at the algorithm level, I reduce computation and memory traffic with sparse attention and low-bit quantization; at the engine level, I design serving systems and schedulers that make full use of GPU and CPU resources. I am now extending this work to agentic AI systems, whose multi-step, tool-using workloads change how inference engines should schedule requests and reuse computation.

Education

PhD	University of Washington , Computer Science and Engineering	Sept 2023 – Now
	• Advisor: Prof. Baris Kasikci and Prof. Arvind Krishnamurthy	
BS	University of Michigan , Computer Engineering	Sept 2021 – Sept 2023
	• Award: University Honors, Dean's List, James B. Angell Scholar	
BS	Shanghai Jiao Tong University , Electrical and Computer Engineering	Sept 2019 – Sept 2021
	• Award: Outstanding Graduate, Undergraduate Excellent Scholarship	

Awards

Allen School Computer Science & Engineering Research Fellowship	Mar 2023
ACM Student Research Competition 1st Place Award (MICRO'22)	Oct 2022
• Presented a poster and gave 10 min talk on micro-architectural implications of Google applications	

Publications

Tactic: Adaptive Sparse Attention with Clustering and Distribution Fitting for Long-context LLMs.	ICLR'26
Kan Zhu , Tian Tang, Qinyu Xu, Zhan Jin, Yile Gu, Zhichen Zeng, Rohan Kadekodi, Liangyu Zhao, Ang Li, Arvind Krishnamurthy, Baris Kasikci	
TraceLab: Characterizing Coding Agent Workloads for LLM Serving.	arXiv'26
Kan Zhu , Mathew Jacob, Chenxi Ma, Yi Pan, Stephanie Wang, Arvind Krishnamurthy, Baris Kasikci	
TeleRAG: Efficient Retrieval-Augmented Generation Inference with Lookahead Retrieval.	MLSys'26
Chien-Yu Lin, Keisuke Kamahori, Yiyu Liu, Xiaoxiang Shi, Madhav Kashyap, Yile Gu, Rulin Shao, Zihao Ye, Kan Zhu , Rohan Kadekodi, Stephanie Wang, Arvind Krishnamurthy, Luis Ceze, Baris Kasikci	
Unleashing Scalable Context Parallelism for Foundation Models Pre-Training via FCP.	MLSys'26
Yilong Zhao, Xiaonan Nie, Kan Zhu , Shuang Ma, Zhichao Lai, Hongxiang Hao, Yang Zhou, Baris Kasikci, Ion Stoica	
Accelerating Large-Scale Reasoning Model Inference with Sparse Self-Speculative Decoding.	MLSys'26
Yilong Zhao, Jiaming Tang, Kan Zhu , Zihao Ye, Chi-Chih Chang, Chaofan Lin, Jongseok Park, Guangxuan Xiao, Mohamed S. Abdelfattah, Mingyu Gao, Baris Kasikci, Song Han, Ion Stoica	

BlendServe: Optimizing Offline Inference with Resource-aware Batching. Yilong Zhao, Shuo Yang, Kan Zhu , Lianmin Zheng, Baris Kasikci, Yifan Qiao, Yang Zhou, Jiarong Xing, Ion Stoica	ASPLOS'26
Themis: Software-Defined Hardware Prefetching. Keisuke Kamahori, Neil Adit, Kan Zhu , Yuqi Mai, Victor Lee, Heiner Litz, Chris Kennelly, Snehasish Kumar, Hanna Alam, Milad Hashemi, David Li, Adrian Sampson, Baris Kasikci, Tipp Moseley, Parthasarathy Ranganathan, Akanksha Jain	arXiv'26
NanoFlow: Towards Optimal Large Language Model Serving Throughput. Kan Zhu , Yufei Gao, Yilong Zhao, Liangyu Zhao, Gefei Zuo, Yile Gu, Dedong Xie, Tian Tang, Qinyu Xu, Zihao Ye, Keisuke Kamahori, Chien-Yu Lin, Ziren Wang, Stephanie Wang, Arvind Krishnamurthy, Baris Kasikci	OSDI'25
PolyServe: Efficient Multi-SLO Serving at Scale. Kan Zhu , Haiyang Shi, Le Xu, Jiaxin Shan, Arvind Krishnamurthy, Baris Kasikci, Liguang Xie	arXiv'25
From Optimal to Practical: Efficient Micro-op Cache Replacement Policies for Data Center Applications. Kan Zhu* , Yilong Zhao*, Yufei Gao, Peter Braun, Tanvir Ahmed Khan, Heiner Litz, Baris Kasikci, Shuwen Deng	HPCA'25
Fiddler: CPU-GPU Orchestration for Fast Inference of Mixture-of-Experts Models. Keisuke Kamahori, Tian Tang, Yile Gu, Kan Zhu , Baris Kasikci	ICLR'25
QUEST: Query-Aware Sparsity for Efficient Long-Context LLM Inference. Jiaming Tang, Yilong Zhao, Kan Zhu , Guangxuan Xiao, Baris Kasikci, Song Han	ICML'24
Atom: Low-Bit Quantization for Efficient and Accurate LLM Serving. Yilong Zhao, Chien-Yu Lin, Kan Zhu , Zihao Ye, Lequn Chen, Size Zheng, Luis Ceze, Arvind Krishnamurthy, Tianqi Chen, Baris Kasikci	MLSys'24
Can Storage Devices be Power Adaptive? Dedong Xie, Theano Stavrinou, Kan Zhu , Simon Peter, Baris Kasikci, Thomas E. Anderson	HotStorage'24

Internships

NVIDIA	Jun 2025 - March 2026
- Enhanced the performance of Megatron-inference on Blackwell GPUs. - Improved the model weight resharding performance for Megatron-RL.	
ByteDance	Feb 2025 - May 2025
Explored efficient scheduling for mixed SLO workloads at large scale serving.	

Open Source Projects

ServingStudio , a workbench for simulating and optimizing LLM serving	ServingStudio 
- Compares configurations with performance models grounded in GPU profiling. - Uses an agent to build and validate top candidates in real frameworks.	
TraceLab , a dataset of real-world coding agent traces	uw-syfi/TraceLab 
- Collected agent sessions and tool calls from Claude Code and Codex. - Integrated into NVIDIA AIPerf for trace replay.	
NanoFlow , a throughput-oriented LLM serving framework	efeslab/Nanoflow 
- Constructed a high-performance serving pipeline using CUDA. - Achieved up to 72% of optimal serving throughput. - Its asynchronous scheduling is adopted in SGLang.	

Talks

NanoFlow: Towards Optimal Large Language Model Serving Throughput.

- Carnegie Mellon University
- ByteDance

Oct 2024
Sept 2024

Research Mentoring

Yufei Gao (THU BS -> UW PhD)

2023-2024

- Overlapping GPU operations with different resource needs

Tian Tang (THU BS -> UW PhD)

2023-2024

- CPU-GPU co-serving MoE models

Qinyu Xu (THU BS -> NYU PhD)

2023-2024

- Clustering based, cumulative attention score targeted sparse attention

Yilong Zhao (SJTU BS -> UCB PhD)

2022-2023

- Activation and weight quantization for LLM

Yuqi Mai (Umich BS -> Cornell PhD)

2022-2023

- Cache prefetcher throttling for Google Traces

Yuewen Hou (Umich BS -> Umich PhD)

2022-2023

- Optimal cache replacement policy for generic cache

Teaching

University of Washington, CSE 599K, LLM Serving Systems

Spring 2025

- Course co-creator
- Co-instructor
- Teaching assistant

Shanghai Jiao Tong University, VG 101, Introduction to Programming.

Summer 2021

- Teaching Assistant for undergraduate course